The Capacity of Absence:
The Conditional Geometry of Perforated Spacetime

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We investigate the effective macroscopic properties of microscopic spacetime perforations. If a void is treated as a simple density deficit, it yields a repulsive Newtonian monopole $M_{\rm eff} = -\rho_0 V$, making complex features such as lacunarity, topology, and fractal dimension gravitationally invisible at leading order. To capture these geometries, we reframe spacetime holes as localized removals of field support. Using the mathematics of perforated-domain homogenization, we formulate macroscopic operators governed by effective accessibility tensors and Newtonian capacity, rather than missing volume alone. For critically scaled families of holes, missing volume can vanish while total capacity remains finite, leaving spectral, screening, and dispersive signatures. We then analyze the associated optical metrics and find that ordinary domain perforation reduces accessibility and focuses waves. Thus anti-gravity-like behavior is not a generic consequence of "missing spacetime", but requires specific microphysical boundary conditions or dispersive sign inversions.

Keywords: perforated spacetime, capacity, anti-gravity, homogenization, spectral dimension, Menger sponge, equivalence principle

I. INTRODUCTION

There is an elementary observation about holes in a positive background that is both correct and disappointing. If a uniform substratum of density $\rho_0 > 0$ fills space, and a bounded region Ω is emptied of that substratum, then the background-subtracted Newtonian source is

$$\rho^{\rm eff}(\boldsymbol{x}) = -\rho_0 \chi_\Omega(\boldsymbol{x}). \tag{1}$$

At distances large compared with the size of Ω, the field is that of a negative monopole,

$$M_{\rm eff} = -\rho_0 V(\Omega). \tag{2}$$

Thus a hole repels, relative to the background. With the convention used below, the potential of the negative effective source is positive, $\Phi = G\rho_0 V/r$, so the acceleration $-\nabla\Phi$ points outward. But this is just the shell theorem and the multipole expansion applied to a subtracted source. It is true in exactly the sense in which a cosmological void is an underdensity relative to the Friedmann–Lemaître–Robertson–Walker (FLRW) mean, and in the same structural sense in which a band hole or a Casimir configuration is defined by subtraction from a reference state. It does not yet explain why a Menger-like or fractal image of spacetime should matter.

The thesis of the present paper is that this disappointment is instructive. The monopole calculation should not be abandoned; it should be promoted to a no-go theorem. If holes are treated merely as missing mass-energy, then their leading exterior field is controlled by missing volume alone. Shape, topology, lacunarity, Hausdorff dimension, and the visual drama of a porous geometry disappear from the far field. Consequently, if the image of "less spacetime" is to have physical content beyond a bookkeeping negative mass, it must enter somewhere else. The natural place is the operator domain on which fields, waves, and geodesics are defined.

The proposed new stream is therefore operator-geometric rather than monopolar. A hole is not primarily a negative source. It is a change in support:

$$\Sigma_\epsilon = \Sigma \setminus H_\epsilon, \tag{3}$$

where Σ is a spatial slice or effective spatial manifold and H_ϵ is a family of unresolved perforations. Fields no longer solve their equations on Σ, but on Σ_ϵ, with boundary conditions on ∂H_ϵ. In a coarse-grained description, the limiting field equation need not be the original equation with an averaged density. Schematic elliptic limits can contain terms of the form

$$-\nabla_i\big(A^{ij}\nabla_j u\big) + \kappa u = f, \tag{4}$$

where A^{ij} is an effective accessibility or tortuosity tensor and κ is a capacity density of the removed set. These two coefficients should not be read as automatically coexisting: in standard perforated-domain homogenization they usually arise in different scaling and boundary-condition regimes.

This is the sense in which absence can have more than volume. A family of tiny holes may have total volume tending to zero while its total Newtonian capacity remains finite. Such holes are invisible to a volume-deficit monopole calculation but visible to the Laplacian, to the spectrum, to wave propagation, and potentially to the effective inertial response of bound modes. This observation is not speculative mathematics; it is the standard lesson of homogenization in perforated domains, where the limiting equation may acquire a zero-order term–the celebrated "strange term coming from nowhere" of Cioranescu and Murat [1].

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The speculative step is to use this mathematics as a new image for spacetime. In Hertz's sense, a physical theory is organized by images whose logical consequences are made to mirror the consequences of nature [2, 3]. The image proposed here is not "negative matter". It is "less support for fields". Its sign is not fixed by that slogan. Ordinary loss of support tends to lower the effective principal tensor, producing a higher optical index and focusing rather than defocusing. Anti-gravity in the operator-geometric sense is therefore a conditional effect, not a generic consequence of perforation. The equivalence principle is not discarded: it is reformulated as the requirement that all freely falling probes couple, in the appropriate regime, to the same coarse-grained principal symbol A^{ij}.

The paper is organized as follows. Section II states the Hertzian image shift. Section III proves the volume-monopole no-go result and explains why the earlier negative-mass formulation is too weak to carry the Menger intuition. Section IV replaces density defects by perforated operator domains. Section V introduces capacity as the first nontrivial invariant of absence and gives a scaling example in which volume vanishes but capacity survives. Section VI reinterprets the Menger image spectrally rather than monopolar. Section VII derives the effective-geodesic interpretation from the principal symbol of the coarse-grained wave operator and analyzes the sign of the effect. Section VIII sketches the interface with inertia and band-hole analogies. Section IX lists observational and theoretical discriminants. Section X collects limitations and open tasks, and Sec. XI summarizes the core image.

II. THE HERTZIAN IMAGE: LESS SPACETIME, NOT NEGATIVE MATTER

The conceptual difficulty with anti-gravity is not only technical. It is pictorial. If one begins with the image of an autonomous negative mass particle, then Bondi's runaway argument is immediate and severe: one has introduced a body with negative inertial mass and negative gravitational charge [4]. If instead one begins with a positive background and removes part of it, the object is not an autonomous particle. It is an absence defined relative to a reference state.

That second image is already familiar in physics. Semiconductor holes are not fundamental positive charges inserted into a band; they are vacancies in a filled electronic background [5, 6]. Casimir energies are not raw vacuum densities; they are differences between two spectral configurations [7, 8]. Cosmological voids are not regions of literal negative matter; they are underdensities relative to the cosmic mean [9–11]. These examples suggest a general methodological rule: an absence can act as an excitation, but only relative to the structure from which it is absent.

The band example also gives a useful warning about sign. A hole's effective mass is not fixed by the mere word "absence"; it is fixed by the curvature of a dispersion relation. Likewise, a spacetime absence need not be repulsive once it is treated as an operator-domain effect. The sign must be derived from the effective operator, not chosen from the image.

The present proposal sharpens this rule. A hole in spacetime should not be represented only by subtracting density from a background. That operation captures the monopole but misses the deeper geometric effect: fields have fewer places to live. The missing set changes domains of operators, boundary conditions, spectra, heat kernels, Green functions, and geodesic accessibility. Thus the slogan is not simply

> holes repel,

but rather

> absence modifies the operator by which fields discover geometry.

This is a useful distinction because it separates three levels that are often conflated:

1. *Density deficit:* the background-subtracted stress-energy has a negative monopole proportional to missing volume.
2. *Support deficit:* the field equations are posed on a perforated domain with new boundary conditions.
3. *Spectral deficit:* the coarse-grained operator has modified principal symbol, capacity terms, and changed spectral dimension.

Only the first is contained in the elementary negative-mass story. The second and third are where a fractal or Menger-type image can become nontrivial.

III. A NO-GO THEOREM FOR DENSITY-ONLY HOLES

We begin with the deliberately standard calculation that makes the older formulation both attractive and too small. Calling it a no-go theorem is a rhetorical use of the multipole expansion: the point is not novelty of the mathematics, but the exclusion of fractal detail from the leading density-defect field.

Theorem 1 (Density-only monopole) *Let $\Omega \subset \mathbb{R}^3$ be bounded and measurable, with finite volume $V(\Omega)$. Remove a uniform positive background density ρ_0 from Ω, and set*

$$\rho^{\mathrm{eff}} = -\rho_0 \chi_\Omega. \tag{5}$$

If the centroid is placed at the origin, then the exterior potential is

$$\Phi(\boldsymbol{r}) = \frac{G\rho_0 V(\Omega)}{r} + \sum_{\ell \geq 2} \frac{1}{r^{\ell+1}} \sum_{m=-\ell}^{\ell} q_{\ell m}(\Omega) Y_{\ell m}(\hat{\boldsymbol{r}}), \tag{6}$$

with $r = |\boldsymbol{r}|$. The leading exterior field is therefore determined only by $V(\Omega)$. All shape information begins at quadrupole order or higher.

Proof 2 *The potential is*

$$\Phi(\boldsymbol{r}) = G\rho_0 \int_\Omega \frac{\mathrm{d}^3 x}{|\boldsymbol{r} - \boldsymbol{x}|}. \tag{7}$$

For r larger than the radius of a sphere containing Ω, the standard spherical-harmonic expansion of $|\boldsymbol{r} - \boldsymbol{x}|^{-1}$ converges absolutely:

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{x}|} = \sum_{\ell=0}^{\infty} \frac{4\pi}{2\ell+1} \frac{|\boldsymbol{x}|^\ell}{r^{\ell+1}} \sum_{m=-\ell}^{\ell} Y^*_{\ell m}(\hat{\boldsymbol{x}}) Y_{\ell m}(\hat{\boldsymbol{r}}). \tag{8}$$

Insertion into Eq. (7) gives the multipole series. The $\ell = 0$ term is $G\rho_0 V(\Omega)/r$, and the $\ell = 1$ term vanishes by the centroid convention.

Corollary 3 *Any two bounded holes of equal volume and common centroid have identical leading $1/r$ exterior potentials in the Newtonian or linearized regime. If both are contained in a ball of radius $\mathcal{R}$, their field difference at $r \gg \mathcal{R}$ is suppressed by $\mathcal{O}(\mathcal{R}^2/r^2)$ relative to the monopole.*

This result is not a defect of the calculation. It is the important message. A sphere, a cube, a multiply connected void, a random foam, and a prefractal Menger construction with the same missing volume produce the same leading density-defect field. Thus the density-only model cannot be the formal home of the Menger intuition. It makes the intuition visually compelling but mathematically inert.

This also clarifies the status of "negative mass" language. At the monopole level one may write

$$M_{\text{eff}} = -\rho_0 V(\Omega). \tag{9}$$

But M_{eff} is not the mass of a fundamental object. It is the coefficient of the $1/r$ term obtained after subtracting a reference configuration. It is therefore a useful far-field bookkeeping parameter, not the core of a new spacetime image.

IV. PERFORATED SPACETIME AS AN OPERATOR DOMAIN

The next step is to distinguish a *source perturbation* from a *domain perturbation*. A source perturbation changes the right-hand side of an equation on a fixed domain:

$$-\Delta u = f + \delta f. \tag{10}$$

A domain perturbation changes where the equation is defined:

$$-\Delta u_\epsilon = f \quad \text{on} \quad \Sigma_\epsilon = \Sigma \setminus H_\epsilon, \tag{11}$$

together with boundary conditions on ∂H_ϵ.

For spacetime holes, Eq. (11) is the more faithful formal image. A missing set is not exhausted by the fact that some energy density has been removed from it. It also imposes a new condition on fields. Depending on the microscopic physics, the boundary of a hole may be absorbing, reflecting, transmitting with a phase shift, or something more exotic. Even without committing to that microphysics, one can study the effective operator produced after coarse graining.

The minimal postulates of the operator-geometric programme are:

1. *Support postulate.* At a microscopic or mesoscopic scale ϵ, the spatial support available to fields is a perforated metric-measure space $(\Sigma_\epsilon, g_\epsilon, \mu_\epsilon)$.

2. *Universal principal-symbol postulate.* In the regime in which free fall is meaningful, all minimally coupled probes share the same leading differential operator, up to species-dependent lower order terms.

3. *Homogenization postulate.* As ϵ becomes unresolved, the operators L_ϵ converge, in an appropriate weak or resolvent sense, to an effective operator L_{eff} on the coarse manifold Σ.

4. *Equivalence postulate.* The effective inertial and gravitational response of freely falling probes is determined by the same coarse-grained principal symbol of L_{eff}.

For a scalar elliptic model, it is useful to write a schematic envelope for possible second-order limits:

$$L_{\text{eff}} u = -\nabla_i (A^{ij} \nabla_j u) + \kappa u. \tag{12}$$

This equation is not meant to assert that $A^{ij} \neq g^{ij}$ and $\kappa \neq 0$ coexist generically. They are different possible operator-level memories of perforation.

a. Finite-fraction principal-symbol regime. If the holes occupy a finite volume fraction, or if the microgeometry acts as a composite medium for transport, the principal part can homogenize to a nontrivial tensor A^{ij}. This tensor records anisotropic accessibility of the perforated domain, much as an effective conductivity tensor records anisotropic transport in a composite. In ordinary obstacle problems this loss of accessibility typically reduces A^{ij} relative to the unperforated value.

b. Critical-capacity strange-term regime. If the holes are Dirichlet obstacles at the critical scaling $a_\epsilon \sim \epsilon^3$ in three dimensions, their total volume vanishes and the principal part remains the original Laplacian to leading order, but a positive lower-order term κu remains. This is the Cioranescu–Murat regime. In the notation of Eq. (4), κ denotes the coefficient γ appearing in the limiting Cioranescu–Murat equation (18). It is not a repulsive force; dimensionally it is a mass-squared or screening term, since capacity density has units length^{-2}.

Thus Eq. (4) should be read as a compact notation for a family of possible coarse-grained remnants. Obtaining both a nontrivial A^{ij} and a nonzero κ in one model would require a mixed or two-scale microgeometry, for example finite-fraction reflecting structure together with critically scaled absorbing holes. This is already a different kind of claim from the negative-monopole story: the field does not repel because a negative mass has been inserted; rather, the coarse-grained geometry redirects propagation because the underlying support has been perforated. The sign of that redirection remains a question for the effective operator.

V. CAPACITY AS THE FIRST NONTRIVIAL INVARIANT OF ABSENCE

Volume is the correct invariant for the monopole of a density defect. It is not the correct invariant for the effect of a small obstacle on a Laplacian. The relevant invariant is capacity.

Definition 4 (Capacity) *For compact $K \subset \mathbb{R}^3$, its Newtonian capacity is*

$$\operatorname{cap}(K) = \inf\Big\{ \int_{\mathbb{R}^3} |\nabla v|^2 \, \mathrm{d}^3x : \; v \in C_c^\infty(\mathbb{R}^3), \\ v \geq 1 \textit{ in a neighbourhood of } K \Big\}. \tag{13}$$

With this normalization, a ball of radius a has $\operatorname{cap}(B_a) = 4\pi a$.

The crucial scaling distinction is

$$V(B_a) = \frac{4\pi}{3}a^3, \qquad \operatorname{cap}(B_a) = 4\pi a. \tag{14}$$

Small holes lose volume as a^3 but retain capacity as a. Hence many very small holes can be negligible as missing volume while remaining finite as missing support.

Proposition 5 (Capacity without volume) *Let a bounded three-dimensional region be partitioned into cells of linear size ϵ, so that the number of cells scales as $N_\epsilon \sim C\epsilon^{-3}$. Put one spherical hole of radius $a_\epsilon = \alpha\epsilon^3$ in each cell. Then the total missing volume tends to zero,*

$$N_\epsilon V(B_{a_\epsilon}) \sim C\epsilon^{-3}\frac{4\pi}{3}\alpha^3\epsilon^9 = \mathcal{O}(\epsilon^6) \to 0, \tag{15}$$

whereas the total capacity has a finite nonzero limit,

$$N_\epsilon \operatorname{cap}(B_{a_\epsilon}) \sim C\epsilon^{-3} 4\pi\alpha\epsilon^3 = 4\pi C\alpha. \tag{16}$$

This elementary estimate is the mathematical heart of the new image. A configuration can be invisible to the density-monopole theorem and visible to the operator. In the critical scaling of perforated-domain homogenization, solutions of

$$-\Delta u_\epsilon = f \quad \text{in} \quad \Omega_\epsilon = \Omega \setminus H_\epsilon, \qquad u_\epsilon = 0 \quad \text{on} \quad \partial H_\epsilon \tag{17}$$

may converge to a solution of

$$-\Delta u + \gamma u = f \quad \text{in} \quad \Omega, \tag{18}$$

where γ is determined by the limiting capacity density of the holes [1, 12]. In the schematic notation of Eq. (4), this is the $\kappa = \gamma$ case with unchanged principal part. The term γu is the canonical example of an operator-level memory of absence. For $\gamma > 0$ it is a screening or mass-squared term: in a static Green-function problem it produces Yukawa-type attenuation, not repulsion.

For the present purpose, Eq. (18) should not be read as a literal gravitational field equation. It is a model of the kind of coarse-grained remnant that holes can produce. Its lesson is that a spacetime defect may be physically meaningful even when its missing volume is too small to generate a significant negative monopole. Its lesson is not that capacity by itself is anti-gravity, but that capacity is a non-volume invariant through which absence can affect spectra and propagation.

VI. MENGER-TYPE ABSENCE AND SPECTRAL GEOMETRY

The Menger sponge is not important because its Hausdorff dimension can be inserted into a Newtonian mass formula. The no-go theorem shows that, at the density-monopole level, such a move is empty. A Menger-type construction is important because it separates geometric invariants that ordinary volume conflates.

At the n-th stage of the standard Menger construction, a cube has been repeatedly perforated. The retained set has volume $(20/27)^n$ times the original volume, tending to zero, while its Hausdorff dimension tends to

$$D_H = \frac{\log 20}{\log 3}. \tag{19}$$

If the removed part is treated only as a mass deficit, the far-field result is determined by the removed volume. But as an operator domain, the same construction has growing boundary complexity, nontrivial connectivity, altered diffusion, and a spectrum not captured by the Lebesgue measure alone.

The broader spectral-geometric question is Kac's: to what extent can one hear the shape of a domain from the spectrum of its Laplacian [13]? Fractal geometries sharpen this question because the heat kernel and density of states may scale with a spectral dimension rather than the embedding dimension [14–16]. For ordinary Euclidean domains the short-time heat trace begins with the volume term, followed by boundary and curvature corrections. For fractal or perforated domains, the boundary and spectral terms can dominate the intuition even when the volume term is simple or vanishing.

Thus the Menger image should be reinterpreted as follows:

> The Menger sponge is not a source with a peculiar mass. It is a test of whether fields experience spacetime through measure alone or through the spectrum of an operator on a perforated support.

In this sense, the visually compelling part of the old picture is saved only after the mass-defect interpretation is demoted.

VII. EFFECTIVE GEODESICS AND THE SIGN PROBLEM

How does an operator-level defect become an anti-gravity-like effect? The bridge is the principal symbol, but the sign is not free. Consider a scalar wave model on a coarse-grained static background,

$$\frac{1}{c^2}\partial_t^2\psi - \nabla_i(A^{ij}\nabla_j\psi) + \kappa\psi = 0. \tag{20}$$

In a high-frequency WKB ansatz $\psi = a\exp(iS/\eta)$, the leading eikonal equation is

$$(\partial_t S)^2 = c^2 A^{ij}\partial_i S\,\partial_j S. \tag{21}$$

The lower-order capacity term κ affects dispersion, amplitude, and bound-state structure, while the principal tensor A^{ij} defines the effective optical geometry. Rays follow the Hamiltonian flow of

$$H(\boldsymbol{x},\boldsymbol{p}) = \frac{c^2}{2}A^{ij}(\boldsymbol{x})p_ip_j. \tag{22}$$

This immediately exposes a sign issue. In the isotropic case $A^{ij} = a(\boldsymbol{x})\delta^{ij}$, Eq. (21) gives

$$|\boldsymbol{p}| = \frac{\omega}{c\sqrt{a(\boldsymbol{x})}}. \tag{23}$$

Thus the effective refractive index is

$$n_{\text{eff}}(\boldsymbol{x}) = a(\boldsymbol{x})^{-1/2}. \tag{24}$$

Ordinary perforation of a propagation medium lowers accessibility: $a(\boldsymbol{x})$ is smaller in the more perforated region. Equation (24) then says that the optical index is larger there. By Fermat's principle rays bend toward the slower, higher-index region. This is focusing, or attraction-like behaviour, not anti-gravity.

a. Worked slab example. Take a two-dimensional slab containing a dilute areal fraction $f(\boldsymbol{x}) \ll 1$ of reflecting circular perforations. For the scalar principal part, the standard dilute composite calculation gives the local transverse effective coefficient

$$a_{\text{eff}}(\boldsymbol{x}) = a_0\frac{1-f(\boldsymbol{x})}{1+f(\boldsymbol{x})} = a_0\big[1-2f(\boldsymbol{x})+\mathcal{O}(f^2)\big]. \tag{25}$$

Consequently

$$n_{\text{eff}}(\boldsymbol{x}) = a_{\text{eff}}(\boldsymbol{x})^{-1/2} = n_0\big[1+f(\boldsymbol{x})+\mathcal{O}(f^2)\big]. \tag{26}$$

For example, let the perforation fraction across the slab be a Gaussian $f(y) = f_0\exp[-y^2/(2\sigma^2)]$. For a ray initially travelling in the x-direction with small slope, Fermat's equation gives

$$\frac{d^2y}{dx^2} \simeq \partial_y \log n_{\text{eff}} = \partial_y f(y) = -\frac{y}{\sigma^2}f(y) + \mathcal{O}(f_0^2). \tag{27}$$

For $y > 0$ this curvature is negative, and for $y < 0$ it is positive: rays bend toward $y = 0$, the most perforated part of the slab. If the perforation fraction is largest along the centre of the slab, then the effective index is also largest there. Fermat rays bend toward that centre: the perforated slab focuses. In this elementary model the operator channel has the attraction-like sign. Defocusing would require the opposite effective-index gradient, for instance microphysics in which the removed structures were scatterers or drag-producing degrees of freedom rather than the propagation support.

Therefore operator-geometric anti-gravity requires an additional sign mechanism. There are three logically distinct possibilities. First, the ordinary density-defect monopole can dominate the motion of massive test bodies, yielding the repulsive field of Theorem 1, while the operator channel produces a separate focusing or dispersive wave effect. Second, the microphysics removed by the hole may be an obstacle, scatterer, drag medium, or polarization cloud rather than the propagation support itself; removing it can increase the effective $a(\boldsymbol{x})$, lower n_{eff}, and defocus rays. Third, more exotic boundary conditions or resonant band-structure effects can invert the effective curvature of the relevant dispersion relation, as happens in ordinary negative effective-mass band physics.

This reconciles, but does not identify, the two pictures in the paper. The monopole picture is a source statement: after background subtraction an underdensity has $M_{\text{eff}} < 0$ and repels at leading Newtonian order. The operator picture is a propagation statement: a perforated support changes the principal symbol and can focus or defocus depending on the sign of the induced effective index. A single microscopic model must show how these two channels combine. If it predicts a repulsive gravitational monopole but a focusing optical principal symbol, the result is not inconsistent; it is a two-channel medium, not a pure metric theory. If one wants a genuine effective spacetime metric seen by all probes, the signs must be made compatible by the microphysics.

The equivalence principle can be expressed cleanly in this language. It does not require that the effective metric equal the original background metric. It requires that all sufficiently ideal test probes couple to the same effective metric. If the same tensor A^{ij} governs the principal propagation of all freely falling matter fields, then local universality of free fall is preserved in the coarse-grained geometry. If different fields see different A^{ij}, or if the density channel and the operator channel give incompatible signs for different probes, the perforated geometry is medium-like and violates universal free fall at the coarse-grained level. This gives the programme a sharp empirical fork.

In weak-field language one may write an effective optical line element

$$ds^2_{\mathrm{eff}} = -c^2\mathrm{d}t^2 + (A^{-1})_{ij}\mathrm{d}x^i\mathrm{d}x^j \tag{28}$$

up to conformal and model-dependent factors. This expression is not intended as a full solution of Einstein's equations. It is the optical metric read from the effective wave operator. The open problem is to derive, from a microscopic spacetime theory, when such an optical metric also acts as the metric governing inertial motion, and when its sign is repulsive rather than attractive.

VIII. INERTIA AS SPECTRAL RESPONSE

The lecture-level motivation behind the present construction is partly the question: what is inertia, and could it be modified? The present stream suggests a conservative answer. Inertia should not be altered by assigning a literal negative mass to a body. It may be altered if the body is a mode whose dispersion relation is changed by the effective operator of the underlying support.

This is precisely how effective mass enters band theory. For a mode with dispersion $E_n(\boldsymbol{k})$, the inverse effective mass tensor is proportional to the Hessian

$$(m_*^{-1})^{ij} = \frac{1}{\hbar^2}\frac{\partial^2 E_n}{\partial k_i \partial k_j}. \tag{29}$$

Near band extrema this tensor may differ greatly from the bare mass and may even have signs that look paradoxical if one forgets that the quasiparticle is a collective excitation of a background. A perforated spacetime image invites the analogous question: can the spectrum of the coarse-grained spacetime operator renormalize the inertial response of localized modes?

Equation (4), read schematically, contains the raw material for such a mechanism. The tensor A^{ij} changes the kinetic term. The capacity density κ shifts the spectral gap or effective mass term. In a periodic or statistically homogeneous microgeometry, one expects Bloch-like or random-medium dispersion relations. In such a setting, "inertia modification" would mean a change in the curvature of the relevant spectral band, not a violation of Newton's second law by a fundamental negative-inertia particle. This is the most concrete way a sign inversion can enter the present programme: not from absence alone, but from the spectrum generated by the absence.

This is also where Sakharov's induced-gravity image becomes suggestive: if gravitational dynamics are emergent from vacuum fluctuations or spectral response of underlying fields, then changing the operator spectrum by changing the support may change the effective geometry itself [17]. The present paper does not derive such a theory. It states the correct mathematical place where such a derivation would have to live: not in $M_{\mathrm{eff}} = -\rho_0 V$, but in the spectral response of fields on perforated support.

IX. OBSERVATIONAL AND THEORETICAL DISCRIMINANTS

The operator-geometric image makes different predictions from the density-only image. The following discriminants are especially useful.

a. Same volume, different capacity. Two regions can have the same missing volume and therefore the same far-field density monopole, while having different capacity densities or boundary structures. The density-only model predicts the same leading Newtonian field. The operator model allows different wave propagation, dispersion, scattering, or inertial response.

b. Zero volume, finite effect. Critical families of holes can have vanishing total volume and finite capacity. Such a configuration is invisible to the old monopole claim but visible to the effective operator. This is the cleanest mathematical signature of the new stream. The expected signature is screening, attenuation, phase shift, or spectral displacement, not automatically repulsion.

c. Principal-symbol versus strange-term regimes. A measured anisotropic ray deflection would point toward a nontrivial A^{ij}, while a mass-like cutoff or Yukawa attenuation would point toward a capacity term κ. Seeing both would be evidence for a mixed or multi-scale microgeometry rather than for the simplest Cioranescu–Murat critical-hole limit.

d. Frequency dependence. A pure Newtonian monopole is not frequency dependent. Operator and spectral effects generally are. Deflection, delay, attenuation, or phase shift that varies with frequency would point toward an effective-medium interpretation rather than a simple negative mass.

e. Equivalence-principle fork. If all probes couple to the same A^{ij}, the effect can be described as a universal effective geometry. If different species or frequencies see different principal symbols, the effect is medium-like and violates universal free fall at the coarse-grained level.

f. Lensing and time delay. The density-only negative monopole suggests defocusing lensing and Shapiro advances rather than delays. The operator-geometric model is more constrained than the earlier wording suggested: ordinary reduction of A^{ij} gives a higher optical index and hence focusing or delay. Defocusing through the optical metric requires A^{ij} to increase in the relevant region, or an effective negative-curvature band response. Additional phase shifts through κ and anisotropic scattering through A^{ij} should not be conflated with monopole repulsion.

g. Void cosmology. Cosmological voids remain the safest empirical analogue of the density subtraction picture. They demonstrate that underdensities can produce repulsive peculiar accelerations relative to the cosmic mean. They do not by themselves demonstrate capacity-driven operator effects. The new programme therefore predicts a conceptual split: voids anchor the monopole story, while laboratory or astrophysical wave-propagation anomalies would be needed to anchor the spectral story. A future model that claims both must verify that the monopole and operator signs are compatible, or else explicitly treat the system as a two-channel medium.

X. DISCUSSION

The old statement "holes repel" is true but too blunt. It identifies only the lowest invariant of absence: missing volume. The new statement is sharper:

> Absence is physically nontrivial when it changes the operator through which fields experience geometry.

This recovers the Menger intuition without pretending that Hausdorff dimension by itself is a gravitational charge. Fractal detail is not visible in the leading Newtonian far field. But it may be visible in capacity, heat-kernel asymptotics, spectral dimension, diffusion, and the principal symbol of the effective wave operator. These are the places where "less spacetime" can become a formal physical hypothesis. They are not, however, automatically places where repulsion appears. Capacity most directly gives screening; reduced accessibility most directly gives focusing. Repulsion requires either the density-defect channel or a derived sign inversion in the operator channel.

The proposal also places clear limits on speculation. It does not construct a negative-mass particle. It does not solve Bondi's runaway problem for fundamental negative-inertia matter. It does not provide an exact solution of the Einstein equations with perforated microscopic support. And it does not claim that Menger-like spacetime structures exist. Its claim is methodological: if one wants to formalize the intuition that holes, gaps, or missing spacetime can matter, the correct variables are not only mass deficit and volume, but capacity, operator domain, spectral dimension, and effective geodesic accessibility.

The slab example above fixes the sign in one elementary regime. At least three mathematical tasks remain. First, one should construct richer families Σ_ϵ and boundary conditions for which the limiting operator can be computed. Second, one should identify which parts of L_{eff} are universal and which are species-dependent. Third, one should determine whether the effective principal symbol can be promoted from an optical metric for selected fields to a genuine metric governing free fall.

XI. CONCLUSION

The core image, des Pudels Kern, is not that a hole has negative mass. That statement is only the shadow of the idea in the Newtonian far field. The core image is that absence changes the support of physical law. Once this is recognized, the apparent triviality of the old calculation becomes useful: it tells us where not to look. The leading monopole sees only volume. The Menger image, if it is to matter, must live in the operator.

The capacity of absence is therefore a candidate organizing variable for a new paper. It connects the physics of holes to rigorous mathematics of perforated domains; it explains how negligible volume can coexist with finite spectral effect; it separates screening, focusing, and defocusing channels; and it preserves the equivalence principle whenever all probes couple to the same coarse-grained principal symbol. In this form, "anti-gravity" is not exotic matter. It is a conditional large-scale geometric effect whose sign must be derived from the microphysics of missing support.

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